trol system provides secure and reliable communication for any type of warship, from patrol boats to submarines. It enables efficient control and management of the ship's overall communication, featuring internal and external communication equipment, remote control with real-time status monitoring and gateway to combat management system. It is composed of the following:

1. Network-centric switches that provide necessary digital and analogue interfaces for non-IP communication devices

2. Multi-functional units to reduce ship's cabling

3. Management terminals for enabling a single operator to fully control and supervise the system, through user-friendly human-machine interface

4. Network access unit for providing digital and analogue interfaces as required by communication system components; responsible for switching and signal distribution functions

TWH-101N by EID marine wireless headset has a compact radio unit and headset, which is used with voice and user terminals. Its main features include:

1. Simultaneous operation of external and internal circuits

2. Encrypted communication with voice/user terminals

3. Up to 64 users (reception)

4. Spread spectrum technology based on 2.4GHz

5. RF output power of -1dBm up to +24dBm

Challenges in designing underwater acoustic and sensor networks

Mohit Kumar Chauhan, manager - pre-sales, Maksat Technologies, explains the challenges in designing underwater acoustic and sensor networks, "Battery power is limited and, usually, batteries cannot be recharged. This is also because solar energy cannot be exploited. Available bandwidth is severely limited, too. Long and variable propagation delays, high bit-error rates, multi-path and fading are other problems. Also, underwater sensors fail because of fouling, corrosion and other such issues."

Fig. 3: Static pressure monitoring on offshore platform (Credit: Yokogawa India Ltd)

Some other challenges are:

1. Underwater sensors are expensive.

2. Underwater network deployment is deemed to be sparse.

3. Underwater network readings are often not correlated due to longer distances between sensors.

4. Higher power is needed in underwater communication, due to longer distances and complex signal processing at receivers.

5. Radio waves do not travel efficiently in salty water. EFY